Prajwal Khanal — Curriculum Vitae

CONTACT INFORMATION

ITC Faculty, University of Twente
Enschede, the Netherlands
prajwalkhanal.earth

p.khanal@utwente.nl
ktm.prajwalkhanal@gmail.com
github.com/prajzwal08 · Google Scholar

RESEARCH INTERESTS

Soil–vegetation–atmosphere interaction; water and carbon fluxes; land surface modelling and deep learning, and where the two can inform each other.

EDUCATION

PhD, Geo-Information Science and Earth Observation *Feb 2023—. . .*

University of Twente, the Netherlands

Modelling soil–vegetation–atmosphere interaction, in both process-based land surface modelling and deep learning approaches to water and carbon fluxes.

M.Sc., Environmental Engineering, Technical University of Munich *Nov 2020—Dec 2022*

Graduated 1.4/5.0 on the German scale, where 1.0 is the highest mark.

Thesis on near-surface versus sub-surface soil moisture in vegetation functioning, published in *Biogeosciences* (2024).

B.E., Civil Engineering, Institute of Engineering, Tribhuvan University, Nepal *Nov 2014—Aug 2018*

Ranked 20th of 192, 77.7%. Thesis on flood inundation mapping of the Babai River.

EMPLOYMENT

PhD Researcher, ITC, University of Twente *Feb 2023—. . .*

WUNDER: ecohydrological responses of agricultural and nature ecosystems in the Netherlands to water use and drought.

Producing field-scale soil moisture and fluxes for agricultural drought monitoring.

Research Intern, Max Planck Institute for Biogeochemistry, Jena *Nov 2020—Feb 2022*

Quantified the relative control of near-surface soil moisture versus terrestrial water storage on global vegetation functioning, from satellite observation alone.

Student Research Assistant, Technical University of Munich *Nov 2021—Mar 2022*

Flood inundation maps for the Ouémé River, Benin (FURIFLOOD) and the Naryn River, Kyrgyzstan (ÖkoFlussPlan), using HEC-RAS 1D unsteady-flow models.

Working Student, Jena-Geos Ingenieurbüro GmbH, Munich *May 2021—Dec 2021*

Excel and VBA tool for greenhouse-gas accounting across stationary combustion, refrigeration and transport.

Civil Engineer (freelance), ECEP, Kathmandu, Nepal *Dec 2018—Dec 2022*

Business plans for community-based water supply projects, for the Department of Water Supply and Sanitation, Government of Nepal.

Smart water supply master plan for the development of Lumbini as a smart city, for the Department of Urban Development and Building Construction.

PUBLICATIONS

Peer-reviewed

Moutahir, H., Sulzer, M., Kiese, R., . . . **Khanal, P.**, et al. (2026). Matching scales of eddy covariance measurements and process-based modelling: assessing spatiotemporal dynamics of carbon and water fluxes in a mixed forest in Southern Germany. *Biogeosciences*, 23, 1719–1738. doi

Zeng, Y., Alidoost, F., Schilperoort, B., . . . **Khanal, P.**, et al. (2025). Towards an open soil-plant digital twin based on the STEMMUS-SCOPE model, following open science. *Computers & Geosciences*, 205, 106013. doi

Khanal, P., Hoek van Dijke, A. J., Schaffhauser, T., et al. (2024). Relevance of near-surface soil moisture vs. terrestrial water storage for global vegetation functioning. *Biogeosciences*, 21(6), 1533–1547. doi

Bhattarai, P., **Khanal, P.**, Tiwari, P., et al. (2019). Flood inundation mapping of Babai basin using HEC-RAS and GIS. *Journal of the Institute of Engineering*, 15(2), 32–44. doi

Under review

Khanal, P., Zeng, Y., Daoud, M. G., Beckers, J., Su, Z. Investigating the impact of multilayer resistance schemes on simulating land surface temperature and turbulent fluxes: evaluation within the STEMMUS-SCOPE model. Preprint on SSRN.

Khanal, P., et al. Deep learning soil moisture and fluxes with multimodal foundation-model embeddings.

Conference contributions

Khanal, P., et al. (2026). Bridging the scale gap: daily 10-metre soil moisture, evapotranspiration and carbon fluxes via multimodal deep learning. *Workshop on Machine Learning for Land and Hydrology*, ECMWF, Reading, UK.

Khanal, P., Hoek van Dijke, A., Zeng, Y., Orth, R. (2023). Near-surface vs. sub-surface soil moisture impacts on vegetation functioning. *EGU General Assembly*, Vienna, EGU23-7670. doi

Khanal, P. (2019). Flood inundation mapping: an inference on the outbreak of flood-borne diseases. *International Conference on Health Engineering in Disaster*.

Awards

Second place, Hackathon on Machine Learning for Earth System Modelling — *Aug 2026*

Quantization in foundation models. Supported by CESOC, ECMWF, TRA Modelling (Bonn), FZ Jülich and the University of Cologne.

Deutschlandstipendium, national merit scholarship, TU Munich — *2021/22*

Institute of Engineering merit-based scholarship, Tribhuvan University — *2014—2018*

Teaching

Teaching Assistant, Technical University of Munich — *2021—2022*

Hydrological and Environmental River Basin Modelling; Integrated Water Resources Management; Rapidly Varying Flow.

Assistant Lecturer, Cosmos College of Management and Technology, Nepal — *Nov 2018—Jan 2021*

Designed and taught an elective on Python and GIS for water resources; tutorials in Fluid Mechanics and Hydraulics.

Software

WUNDER data viewer, ITC, University of Twente — *2026*

Interactive app for the WUNDER weather and soil moisture loggers, so growers at the food forests can follow drought as it develops. App · Source

Skills

Modelling: STEMMUS-SCOPE, land surface modelling, HEC-RAS 1D unsteady flow
Programming: Python, deep learning frameworks, GIS, Excel/VBA
Data: eddy covariance and flux-tower data, satellite soil moisture, ICOS
Languages: English, Nepali